UChicago REU Notes Week 4

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Contents

Remark 0.1. Assume everything is compactly generated, so we have a natural homeomorphism $\mathcal{U}(X \times Y, Z) \cong \mathcal{U}(X, \mathcal{T}(Y, Z))$, and $\mathcal{T}(X \wedge Y, Z) \cong \mathcal{T}(X, \mathcal{T}(Y, Z))$. Here, $\mathcal{U}$ are compactly generated spaces, $\mathcal{T}$ are based spaces.

For basic reviews of G-spaces, I'll type up the notes later.

Here, we also assume V is a finite dimensional $\mathbb{R}$-inner product space with a G-representation.

1 Basics of Equivariant Homotopy

2 Motivation of Categorical Aspect

Goal: Understand to construct genuine G-spectra. A naive G-spectra is a spectra with a G-action (suposedly the structure maps are all G-equivariant + homotopy equivalence, or homeomorphism).Seems like we require extra structures on the spectra. One of the distinction seems to be: naive G-spectra represents $\mathbb{Z}$-graded cohomology theories, genuine G-spectra are used to represents cohomology theories for real representations (so it's also encoded using representations). I particular, understanding what is the equivariant infinite loop space machine (construct from G-space / category, forming a G-spectra).

Definition 2.1. Given a category $\mathcal{V}$, $X \in \mathcal{V}$ is a G-object if it has a group homomorphism $G \to \mathrm{Aut}(X)$.

$\mathcal{V}_G$ denotes the category of G-objects, with morphisms being in $\mathcal{V}$.Each $\mathcal{V}(X, Y)$ has a natural G-action by conjugation, where $g \in G$ acts as $f \mapsto g \circ f \circ g^{-1}$.

$G\mathcal{V} \subseteq \mathcal{V}_G$ denotes the subcategory where homsets are limited to G-maps (so $G\mathcal{V}(X, Y) := \mathcal{V}(X, Y)^G$, fixed points of the homsets). Another notation is $G\mathcal{V} := (\mathcal{V}_G)^G$.

Definition 2.2. Given $X \in G\mathcal{T}$ (based G-space), assume inclusion of basepoint $* \to X$ is nondegenerated in the G-case, meaning it's a G-cofibration (satisfying G-homotopy extension property for G-equivariant maps, abbreviated as G-maps). In particular, all the homotopies in the extension property is required to be a G-map at all time t (another way of phrasing, interval I has trivial action, so action at slice $X \times t$ is equivalent to action on $X \times I$).

By chasing through the homotoopy extension diagram (by restricting all homotopies to be G-homotopy), then inclusion of basepoint $* \to X^H$ becomes a cofibration with respect to G-maps (Note: Not necessarily a G-cofibration, because in general X^H is not closed under G-action). However, since all G-space is naturally an H-space (so is homotopy), then we can understand this as "H-cofibration", as H acts trivially on X^H :0

Definition 2.3. A G-map $f : X \to Y$ is a weak equivalence of G-spaces (or G-equivalence), if the fixed point maps $f^H : X^H \to Y^H$ are weak homotopy equivalences for all subgroups $H \leq G$.

Similar G-whiteheads theorem states if X, Y are G-CW complexes, then f is also a G-homotopy equivalence.

Definition 2.4. A small category $\mathcal{C}$ is a *topological category*, if the set of objects Ob and set of all morphisms Mor are topological spaces, such that the following four maps are continuous:

- Identity map $i : \mathrm{Ob} \to \mathrm{Mor}$ by $i(x) := \mathrm{id}_x$ is a continuous map.
- Source map $s : \mathrm{Mor} \to \mathrm{Ob}$ by $s(f : X \to Y) := X$ is continuous.
- Target map $t : \mathrm{Mor} \to \mathrm{Ob}$ by $t(f : X \to Y) := Y$ is continuous.
- Composition $\circ : \mathrm{Mor} \times_{t,s} \mathrm{Mor} \to \mathrm{Mor}$ is continuous.

Which, the last one is collecting all pair (X, Y) with morphism $f : X \to Y$ between them.

In particular, s, t are retracts of i, showing i is an embedding. (check https://mathoverflow.net/questions/85717/segals-original-definition-of-a-topological-category)

There is another notion enriched over $\mathcal{T}$.

Notation 2.5. Let Cat denotes the category of small topological categories, which Cat_G denotes the topological G-categories (meaning G acts on both $\mathrm{Ob}, \mathrm{Mor}$ such that $i, s, t, \circ$ are all G-maps), an equivalent formulation is a group homomorphism $G \to \mathrm{Aut}_{\mathsf{Cat}}(\mathcal{C}, \mathcal{C})$, by checking the definition.

Part of the reason we endow this definition, is so that for subgroup $H \leq G$, the *H-fixed category* $\mathcal{C}^H$ will make sense, namely we need the action on $\mathrm{Ob}, \mathrm{Mor}$ to be compatible.

Remark 2.6. Given the classifying space functor B, it realizes any topological category $\mathcal{C}$ into $B\mathcal{C}$.

Question: Whatare the comparison between the two spaces?

Which, for some reason it's product preserving, and it sends objects in Cat to $\mathcal{U}$, $G\mathsf{Cat}$ to $G\mathcal{U}$. In particular, it preserves the operad action (as preserving product also preserves the action).

3 E_V Operads, V-fold Loop G-Spaces

3.1 Little Disks / Steiner Operad

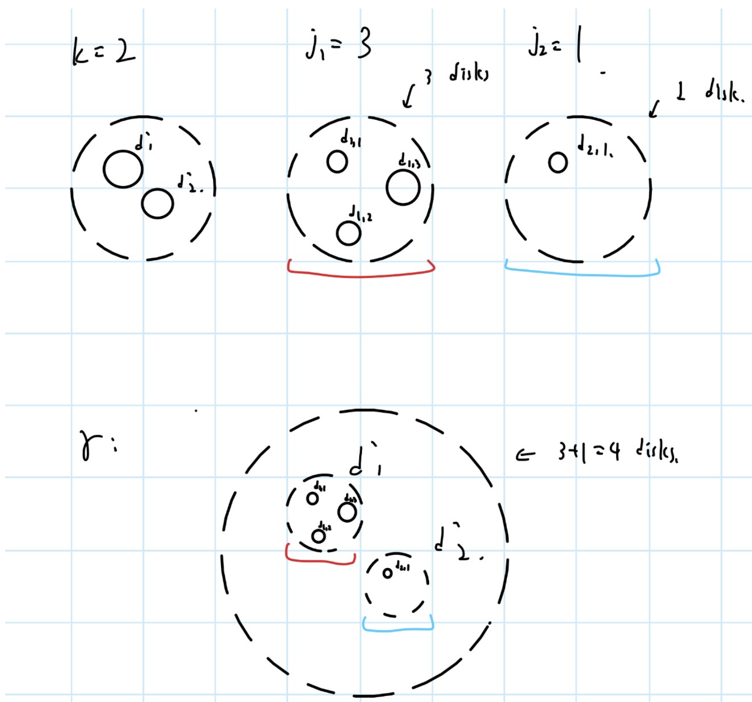
Definition 3.1. Given $D(V) \subset V$ the open unit disk (all $v \in V$ with $\|v\| < 1$), a *little V-disk* is an affine map $d : D(V) \to D(V)$, meaning all $d(v) = rv + v_0$, where $r \in [0, 1)$ and $v_0 \in D(V)$ represents the radius and the center.

Define group action $g \in G$ on collection $(gd)(v) := rv + gv_0$, which forms an evident G-action with restriction of action $G \to O(V)$ (the real unitary group of V).

Define the *little disk operad* by definig $\mathcal{D}_V(j) := \{(d_1, ..., d_j) \mid d_i \text{ shares empty image intersection}\}$. Geometrically, it's all the possible arrangment of j non-overlapping disks in $D(V)$. Define the evident structure map $\gamma : \mathcal{D}_V(k) \times \mathcal{D}_V(j_1) \times ... \times \mathcal{D}_V(j_k) \to \mathcal{D}_V(j)$ as follow:

$$\gamma((d'_1, ..., d'_k); (d_{1,1}, ..., d_{1,j_1}), ..., (d_{j_k,1}, ..., d_{j_k,j_k})) = ((d'_1 \circ d_{1,1}, ..., d'_1 \circ d_{1,j_1}); ...; (d'_k \circ d_{j_k,1}, ..., d'_k \circ d_{j_k,j_k}))$$

Geometrically, it's as follow (for an example):



This is obvious a Σ-operad (as we defined it to be function composition). There is a correspondance of this definition to configuration space: Let $F(V, j) \subset V^j$ be the ordered configuration space of j-tuples of distinct points of V, with G acts on V^j diagonally (which restricts to an action on $F(V, j)$).

Theorem 3.2. *There is a $(G \times \Sigma_j)$-homotopy equivalence $\mathcal{D}_V(j) \simeq F(V, j)$ for all $j \geq 0$.*

More explicitly, rescaling V to $D(V)$ (by a map $[0, \infty) \to [0, 1)$) so that $F(V, j) \cong F(D(V), j)$, then send $(d_1, ..., d_j)$ to the centers $(v_1, ..., v_j)$, and conversely given $(v_1, ..., v_j)$, choose r to be half the smallest distance between these points, then send to the corresponding $(d_1, ..., d_j)$ (each d_i with radius r and center d_i.

Which, since there is a free $(G \times \Sigma_j)$-action (the space is not contractible or a local equivalence), so this is called an E_V-operad.

Definition 3.3. $\mathcal{C}_G$ an *operad of G-spaces* is an *E_V-operad* if there is a morphism of operads $\mathcal{C}_G \to \mathcal{D}_V$, such that each level $\mathcal{C}_G(j) \to \mathcal{D}_V(j)$ is a weak equivalence.

Which, given $V \subset W$ two G-orthogonal representations, in general there's no morphism of operad $\mathcal{D}_V \to \mathcal{D}_W$ that's compatible with suspension. (For details, the non-equivariant version has a suspension map between little n-cube operad to $(n+1)$-cube operad, which is compatible with suspension; hence, taking the colimit results in an E_∞-operad. In this case it doesn't work).

Definition 3.4. Let E_V be the space of embeddings $V \to V$ (not necessarily G-embeddings), together with G acting on E_V with conjugation.

Define $\mathrm{Emb}_V(j) \subset E_V^j$be the G-subspace with ordered$j$-tuples of embeddings $(\gamma_1, ..., \gamma_j)$,such that each $i \neq j$ has disjoint image. The collection Emb_V naturally forms an operad in $G\mathcal{U}$, by the existence of identity embedding, and similar notion of "composite embedding" as little V-disk operad.

Definition 3.5. Define $R_V \subset E_V = \mathrm{Emb}_V(1)$ to be all distance-reducing embeddings, i.e. $f : V \hookrightarrow V$ such that $\|f(v) - f(w)\| \leq \|v - w\|$ for all $v, w \in V$. (Which, composition is closed).

Define a *Steiner path* $h : I \to R_V$ so that $h(1) = \mathrm{id}_V$. Let P_V denotes the path space of Steiner paths, define $\pi : T_V \to R_V$ by $\pi(h) = h(0)$.

Define $\mathcal{K}_V(j)$ to be G-space of ordered j-tuples of Steiner paths $(h_1, ..., h_j)$, such that each $\pi(h_i)$ has disjoint image, with unit in $\mathcal{K}_V(1)$ being the constant Steiner path. Define structure map $\gamma : \mathcal{K}_V(k) \times \mathcal{K}_V(j_1) \times ... \times \mathcal{K}_V(j_k) \to \mathcal{K}_V(j)$ as follow: given $(h'_1, ..., h'_k) \in \mathcal{K}_V(k)$, and $(h_{i1}, ..., h_{ij_i})$, we define $h_{il} \cdot h'_i : I \to R_V$ by sending t to $h_{il}(t) \circ h'_i(t)$, which it's still a Steiner path. And, composition of such embedding preserves disjoint images, so this defines a map of operad by writing similar structure map.

There is a natural map of operad $\pi : \mathcal{K}_V \to \mathrm{Emb}_V$ (by evaluating each Steiner path at 0). There is another map of "evaluation" at $\mathbf{0} \in V$, providing a $(G \times \Sigma_j)$-map $c : \mathrm{Emb}_V(j) \to F(V, j)$.

Theorem 3.6. *There is an equivalence of operad $\iota : \mathcal{D}_V \to \mathcal{K}_V$.*

Intuitively, this can be thought of as: Embedding of disks is equivalent to embedding of V (by rescaling $D(V)$ back to the original space).